ENERGY FUTURES STRATEGY · QUANTITATIVE RESEARCH

ATR Butterfly Analysis — Final Report

CL (WTI) & LCO (Brent) · Adjacent-month butterfly spreads · ATR(14) close-only · Regime-stratified · 2021–2026

PEAK LCO ATR
0.449
LCO M2 fly · Deep-Backwardation

PEAK CL ATR
0.159
CL M2 fly · Deep-Backwardation

FRONT-END / BACK-END PREMIUM
9.8×
LCO M2 vs M8 fly in Deep-Back

CALMEST REGIME
Stable-Dep.
LCO M8 ATR = 0.013 (floor)

Core insight: ATR in butterfly spreads is heavily regime-dependent and decays steeply along the fly ladder. LCO M2 fly volatility in **Deep-Backwardation** (ATR 0.449, P90 = 1.078) is **9.8× higher** than the M8 fly in the same regime (0.046), and **34× higher** than in Stable-Depressed (0.013). The current strategy treats all positions as equal risk — it does not.

CL (WTI) — ATR BY REGIME × FLY

Regime	M2 Fly	M6 Fly	M7 Fly	M2/M7 Ratio
Backwardation-Deficit	0.0519	0.0140	0.0133	3.9×
Contango-Surplus	0.0331	0.0134	0.0135	2.5×
Deep-Backwardation	0.1592	0.0361	0.0265	6.0×
Deep-Contango	0.0315	0.0191	0.0200	1.6×
Easing-Backwardation	0.0477	0.0155	0.0151	3.2×
Easing-Contango	0.0250	0.0121	0.0116	2.2×
Stable-Depressed	0.0215	0.0103	0.0106	2.0×
Stable-Elevated	0.0559	0.0145	0.0140	4.0×
Transition-Tightening	0.0504	0.0134	0.0137	3.7×

Low ATR — High ATR

Note: CL M3/M4/M5 flies excluded — CL M4 contract has only 357/1920 trading days (severely sparse). M2/M7 ratio quantifies front-end stress vs calm back-end in each regime.

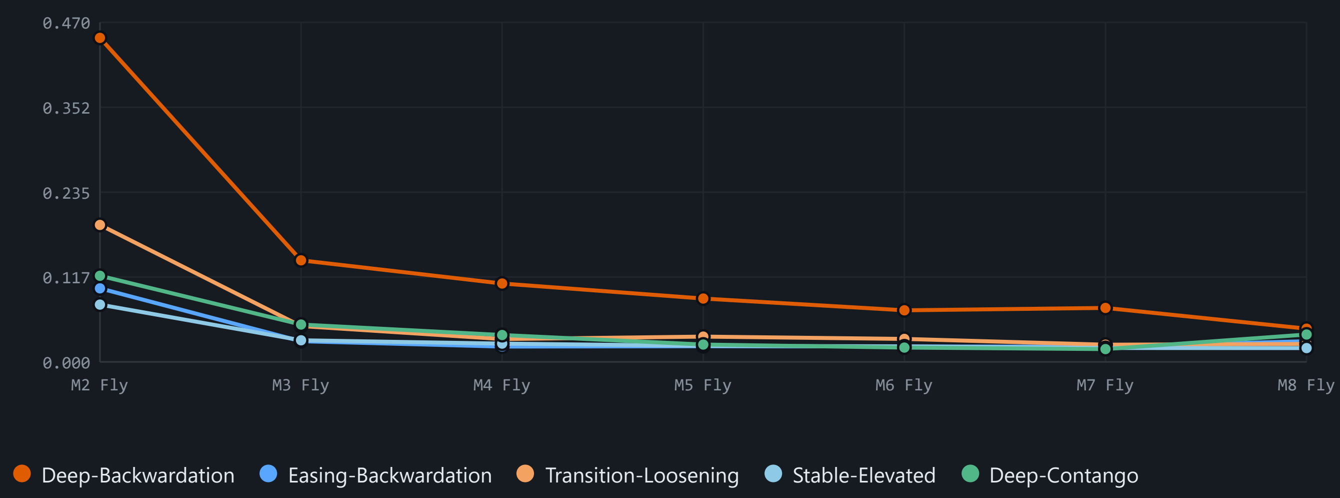
LCO (BRENT) — ATR BY REGIME × FLY

Regime	M2	M3	M4	M5	M6	M7	M8	M2/M8 Ratio
Backwardation-Deficit	0.0906	0.0355	0.0277	0.0277	0.0263	0.0203	0.0190	4.8×
Contango-Surplus	0.0614	0.0235	0.0221	0.0193	0.0176	0.0209	0.0214	2.9×
Deep-Backwardation	0.4486	0.1406	0.1085	0.0877	0.0714	0.0747	0.0460	9.8×

Regime	M2	M3	M4	M5	M6	M7	M8	M2/M8 Ratio
Deep-Contango	0.1192	0.0517	0.0372	0.0240	0.0198	0.0177	0.0379	3.1×
Easing-Backwardation	0.1018	0.0286	0.0212	0.0217	0.0213	0.0211	0.0286	3.6×
Stable-Depressed	0.0945	0.0186	0.0215	0.0163	0.0165	0.0158	0.0128	7.4×
Stable-Elevated	0.0793	0.0299	0.0253	0.0222	0.0215	0.0192	0.0191	4.2×
Transition-Loosening	0.1896	0.0496	0.0316	0.0350	0.0318	0.0238	0.0248	7.6×
Transition-Tightening	0.1134	0.0384	0.0282	0.0345	0.0282	0.0176	0.0187	6.1×

Low ATR High ATR

LCO ATR TERM STRUCTURE — FLY LADDER BY REGIME



CL VS LCO M2 FLY ATR — CROSS-PRODUCT COMPARISON

Regime	CL M2 ATR	LCO M2 ATR	LCO/CL Ratio	Interpretation
Deep-Backwardation	0.1592	0.4486	2.82×	Brent front elevated — structural premium persists
Easing-Backwardation	0.0477	0.1018	2.13×	Brent front elevated — structural premium persists
Stable-Elevated	0.0559	0.0793	1.42×	Near-parity — WTI/Brent dynamics in sync
Stable-Depressed	0.0215	0.0945	4.40×	Brent strongly amplified — North Sea stress
Backwardation-Deficit	0.0519	0.0906	1.75×	Mild Brent premium — CL/LCO broadly aligned
Deep-Contango	0.0315	0.1192	3.78×	Brent strongly amplified — North Sea stress
Contango-Surplus	0.0331	0.0614	1.85×	Mild Brent premium — CL/LCO broadly aligned

STRATEGY INTEGRATION — 5 ACTIONABLE ENHANCEMENTS

01 · SIZING **ATR-Adjusted Position Sizing**

02 · FILTER **ATR Entry Filter — Trade Only Active Flies**

Replace fixed ±1 position with risk-normalized size. Every fly trade then risks the same dollar amount per day regardless of which fly or which regime. In Deep-Backwardation, LCO M2 fly ATR (0.449) is 9.8× the M8 fly — the current fixed-weight scheme unknowingly takes 9.8× more risk on M2.

```
risk_budget <- 0.002  # 0.2% of notional per day target

# Compute rolling ATR for each instrument
oos[, atr_m2fly := atr(lco_m2fly, n = 14)]

# Size = signal × (risk_budget / ATR), capped at ±1
oos[, sz_m2fly := fifelse(
  !is.na(atr_m2fly) & atr_m2fly > 0,
  pmin(1, risk_budget / atr_m2fly),
  1.0
)]
oos[, pos_lco_m2fly_sized := raw_sig * sz_m2fly]
```

IMPACT: HIGH EFFORT: MEDIUM

Only enter a fly position when ATR is above the 25th percentile of its historical distribution for that product. When ATR is extremely low (<P25), the fly is "stuck" — carry-only, no directional move to capture. Signal fires but there's nothing to extract.

```
atr_data <- fread("phase3c/atr_flies_daily.csv")

# Compute rolling 252-day ATR percentile
atr_data[, atr_pct := frank(atr, ties.method="average") /
           .N, by = .(product, fly)]

# Merge and filter: require ATR > P25 to trade
oos <- merge(oos, atr_data[fly=="m2" & product=="LCO",
              .(date, atr_pct_m2 = atr_pct)], by="date")
oos[, pos_lco_m2fly_filtered :=
      fifelse(atr_pct_m2 > 0.25, pos_lco_m2fly, 0L)]
```

IMPACT: MEDIUM EFFORT: LOW

03 · SIGNAL M2/M6 ATR Ratio — Curve Stress Indicator

The ratio of M2-fly ATR to M6-fly ATR measures front-end stress relative to back-end. In Deep-Backwardation, CL M2/M6 = 4.4× and LCO M2/M6 = 6.3×. When this ratio spikes while the regime classifier still says "Easing-Backwardation", the physical market is leading the model — use it as a regime transition early warning with a 1–3 day lead.

```
cl_atr <- fread("phase3c/atr_flies_daily.csv")
cl_sub <- cl_atr[product == "CL"]
cl_wide <- dcast(cl_sub, date ~ fly,
                 value.var = "atr")
cl_wide[, stress_ratio := m2 / pmax(m6, 1e-6)]

# Rolling 20-day z-score of stress ratio
cl_wide[, stress_z := (stress_ratio -
  frollmean(stress_ratio, 20, na.rm=TRUE)) /
  frollapply(stress_ratio, 20, sd, na.rm=TRUE)]

# Flag: stress_z > 1.5 = potential regime transition
cl_wide[, regime_stress_flag := stress_z > 1.5]
```

IMPACT: HIGH EFFORT: MEDIUM

04 · THRESHOLD ATR-Scaled Signal Threshold

Replace the static thresholds (0.04 / 0.10) with ATR-scaled ones. In high-ATR environments the spread is noisier — require a stronger signal before entering. In low-ATR environments (cleaner trends), lower the bar. This reduces false signals in volatile regimes and captures more in calm ones.

```
# Base threshold scaled by normalized ATR
# ATR_norm = ATR / rolling_median(ATR, 60)
oos[, atr_norm_m2 := atr_m2fly /
  frollapply(atr_m2fly, 60, median, na.rm=TRUE)]

# Adaptive threshold: base × (0.5 + 0.5 × ATR_norm)
# → doubles at 3× median ATR; halves at very low ATR
BASE_THR <- 0.04
oos[, adaptive_thr := BASE_THR *
  pmin(2, pmax(0.5, 0.5 + 0.5 * atr_norm_m2))]

oos[, pos_adaptive :=
  fifelse(sig > adaptive_thr,  1L,
  fifelse(sig < -adaptive_thr, -1L, 0L))]
```

IMPACT: MEDIUM EFFORT: MEDIUM

05 · CROSS-PRODUCT CL vs LCO M2 ATR Basis — Brent-Specific Stress Signal

When LCO M2 fly ATR spikes but CL M2 fly ATR stays flat, it's a Brent-specific event (North Sea loading disruption, OPEC+ cut affecting light crude). The reverse (CL spike, LCO flat) is WTI-specific (Cushing inventory, pipeline capacity). This basis can be traded as a cross-product fly divergence or used as a regime confirmation signal for which product's signal to trust more.

Cross-product ATR ratio in key regimes (from current data):

Regime	CL M2	LCO M2	LCO/CL	Read
Deep-Backwardation	0.159	0.449	2.82×	Brent front amplified — North Sea stress
Easing-Backwardation	0.048	0.102	2.13×	Brent still elevated — structural premium
Stable-Elevated	0.056	0.079	1.41×	Near-parity — WTI/Brent dynamics aligned
Stable-Depressed	0.022	0.095	4.32×	Brent front volatile despite low regime — divergence

```
atr_wide <- dcast(atr_data, date ~ paste0(product, "_", fly),
                  value.var = "atr")
# CL and LCO M2 fly ATR basis
atr_wide[, cross_basis := lco_m2 / pmax(CL_m2, 1e-6)]

# z-score: spike = Brent-specific stress
atr_wide[, cross_z := (cross_basis -
  frollmean(cross_basis, 40, na.rm=TRUE)) /
  frollapply(cross_basis, 40, sd, na.rm=TRUE)]

# Signal: if cross_z > 2 → over-weight LCO, under-weight CL
# if cross_z < -2 → opposite
```

IMPACT: HIGH EFFORT: HIGH

IMPLEMENTATION PRIORITY

#	Enhancement	Impact	Effort	What it fixes
01	ATR Position Sizing	HIGH	MEDIUM	Equalizes dollar risk across all fly/regime combos
02	ATR Entry Filter	MEDIUM	LOW	Skips dead flies — removes low-quality trade noise
03	M2/M6 Stress Ratio	HIGH	MEDIUM	Early warning 1–3 days before regime transition
04	Adaptive Threshold	MEDIUM	MEDIUM	Reduces false entries in high-ATR / noisy environments
05	Cross-Product ATR Basis	HIGH	HIGH	Identifies Brent-specific vs WTI-specific stress events